Marcus Yoo

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EDUCATION

Rose-Hulman Institute of Technology **May 2028**

Bachelors of Science, Mechanical Engineering *Terre Haute, IN*

Related Coursework: Fluids, System Dynamics, Thermodynamics, Statics II, Design for Manufacture

INTERNSHIP EXPERIENCE

Boeing Co. **June 2023 - August 2024**

Flight Performance Engineering Intern *Huntington Beach, CA*

- Conducted simulations for takeoff and landing speeds using advanced proprietary models; results directly contributed to operational manuals, ensuring accurate guidance for flight crew across more than 100 unique customer configurations
- Developed python programs to batch process fluid simulation data, boosting operational efficiency, cutting down human time spent processing each simulation by 15%
- Manufactured composite and thermal protection systems test coupons for empirical property verification through X-ray photoelectron spectroscopy, contributing to spacecraft and aircraft coatings development

ADDITIONAL EXPERIENCES

Rose Rocketry **August 2024 - May 2027**

NASA Student Launch Member *Terre Haute, IN*

- Drafted and refined project proposals and technical reports, effectively communicating project objectives and progress
- Developed a mechanical fin attachment system to minimize breakage, increasing ease of rapid design iteration
- Pursuing NAR L2 Certification through design, testing, and flying a fiberglass rocket with an interchangeable payload bay

Sage Hill Robotics (FRC Team 5835) **August 2020 - July 2024**

Team Captain | August 2023 - July 2024 *Newport Coast, CA*

- Collaborated with teammates to rebuild and expand the FIRST robotics team from 4 members to over 40, overcoming the challenges posed by the COVID-19 pandemic, introducing over 60 Sage Hill School students to fields of engineering
- Guided team to 8/48 teams at the Orange County Regional Competition, demonstrating strategic planning and leadership
- Managed a $19,000 budget, overseeing accounting and finance to ensure the team had the necessary resources
- Implemented project management techniques to coordinate group efforts, resulting in a cohesive and efficient team

Design Lead | August 2020 - August 2023

- Mentored group of 2-6 members, developing and enhancing the team's solution seeking workflow
- Applied creative design ethos, keeping in mind manufacturing limitations and identifying possible workarounds

Rose-Hulman Art Curator's Office **September 2025 - June 2027**

Assistant Art Curator *Terre Haute, IN*

- Curated and installed exhibitions across campus spaces, enhancing the daily environment for 2,000+ students and faculty while providing recognition for emerging local artists.
- Coordinated with faculty to align art placement with community needs, cultivating a more vibrant academic culture.

Rose-Hulman YDSA **February 2026 - May 2027**

Co-founder and Vice Chair *Terre Haute, IN*

- Co-founded the chapter, recruiting 30+ members and building the founding community through outreach across campus
- Lead weekly discussions on labor & the engineer's role in the workforce, fostering a forum for socioeconomic dialogue

CERTIFICATIONS & SKILLS

- **Certifications:** Cisco Level II Network Technician; Audinate Dante Level III Certification; ASA Keelboat Certification
- **Adobe Products:** Illustrator; FrameMaker, Indesign; Premiere; After Effects; Photoshop; Lightroom
- **Design & Simulation**: Solidworks & Catia; Autodesk Fusion 360; Satellite Tool Kit; Ansys Fluent; KiCAD; Blender
- **Programming:** Java; Python; C++; Matlab

INTERESTS & ACTIVITIES

- **Interests:** Aviation; Space; Rocket Propulsion; Outdoor/Adventure; Robotics & Automation; Prosthetics & Orthotics
- **Activities:** Climbing, Photography & Darkroom; Ceramics

ROCK CLIMBING

- **Notable Ascents:** Depression v10-, Attack of the Staph v9, The Cackler v9, Year of the Cat v9, Jungle Book v8
- **Rose-Hulman Climbing Team Manager:** Directed training curriculum and travel logistics for 15+ climbers, driving two first-place finishes and a 3rd overall team ranking at the 2025 Collegiate National Championships
- **RH Student Rec. Center Routesetter:** Set commercial-style boulder problems and managed wall upkeep, hold inventory, and equipment purchasing for the Rose-Hulman climbing community
- **Youth Climbing Coach:** Coached youth climbers on technique and safety at Sender One Aliso Viejo, managing parent expectations and progress communication

PROJECT SHOWCASE

PCBrule: A PCB-based Analog Pocket Calculator **July 2025 - March 2027**

Personal Product Design Project

Developing a slide rule from printed circuit board material, modeled after the Pickett N600-ES, the slide rule Apollo astronauts took to the moon. I wanted to build a product that was both functional and beautiful, straddling the gap between artistry and engineering, this project is the thesis of my engineering philosophy. Developing the design and preparing it for larger-scale manufacture has been a challenge, and it has shown me what the product development process is really about. This project drew on skills from many corners: PCB & CAD modeling, GD&T, and graphic design.

Learn more on my website here: marcusjyoo.com/pcbrule

Rose-Rocketry: Mechanical Fin Attachment System **August 2024 - April 2025**

NASA Student Launch: 2024-25 Vehicle Subassembly

Designed and built a mechanical fin attachment system for the Project Kline subscale vehicle. In prior seasons, fin damage required grinding, filing, and sanding down epoxy fillets before reattachment, a hazardous 24+ hour process. My system reduced replacement to a 30-minute hot-swap, letting the team iterate on fin geometries at the launch site. The design proved effective enough that the full-scale vehicle adopted a variant of it, validating the mechanical approach beyond the subscale platform. The project taught me that designing for serviceability is often worth the added complexity in development hardware.

Homebuilt Snowskate Project **January 2026 - April 2026**

Personal Fabrication Project

Built a snowskate from a thrifted kids' ski and plastic skateboard deck. Designed rocker-style mounting trucks in SolidWorks for waterjet cutting. The rocker geometry lets the ski flex naturally under the deck instead of being overconstrained, preserving the camber that governs stiffness. Sizing the truck stack height and mounting pattern required balancing deck clearance against the ski's flex envelope, and the waterjet process constrained every feature to a 2D profile.